Janko Štrkalj

Backend & Software Engineer

CONTACT

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SKILLS

- Python (FastAPI)
- PHP (Laravel), JavaScript, HTML, CSS
- MySQL, SQLite, InfluxDB
- MQTT, IoT & Embedded Systems (Raspberry Pi)
- AI, ML, Git, REST APIs, LLM APIs, Linux

ORGANIZATIONS

Youth Club SKC
02/2025 - Present | Kragujevac, Serbia
Member of Youth Club. Actively involved in student cultural activities, events, and community initiatives.

Neutrinos IT
01/2025 - Present | Kragujevac, Serbia

PROFILE

Recently graduated Software Engineer with hands-on experience across full-stack web development, IoT/embedded systems, and AI-assisted automation. Delivered two client projects as a freelancer, from database design to deployment, with a growing focus on backend systems and applied AI.

PROFESSIONAL EXPERIENCE

Freelance Software Engineer

07/2025 - 06/2026 | Remote

Neighborhood Energy Orchestrator - Dutch energy-tech client (03/2026 - 06/2026)

- Built an IoT proof of concept for a neighborhood energy-orchestration platform, part of an innovation project funded by the European Regional Development Fund (EFRO).
- Developed edge software for two Raspberry Pi 4 devices and a FastAPI cloud orchestrator that coordinates solar panels, batteries, EV chargers and heat pumps to reduce grid congestion.
- Designed and implemented a simulation engine modelling hourly per-household consumption for 8- and 30-house districts under 100 kW and 700 kW transformer limits; with the orchestrator enabled, the system met the project's KPI targets of ≥15% lower transformer peak load and ≥20% higher self-consumption.

GR Centric - GRC platform (07/2025 - 12/2025)

- Built a full-stack compliance-automation platform end-to-end in collaboration with a UI/UX designer, covering the PHP/MySQL backend, database design and front-end implementation.
- Developed a Python background worker that processes queued document jobs from MySQL, with a lock against concurrent runs, connecting the platform to an AI pipeline that extracts security controls from framework documents and generates summaries; moved from a local Llama model to GPT-4.1 for better quality at acceptable cost.

IT Support - Eurobank Direktna

12/2023 - 03/2025 | Kragujevac, Serbia

Day-to-day technical support for internal bank teams: software installation and troubleshooting of workstations, printers and banking systems.

Internship - Comtrade - EDIT

07/2023 - 07/2023 | Kragujevac, Serbia

Two-week team internship: designed and deployed a cloud-native app.

EDUCATION

Bachelor of Engineering, Software Engineering - Faculty of Engineering

10/2019 - 09/2026 | Kragujevac, Serbia

Specialized Program in Mathematics - First Kragujevac Gymnasium

09/2013 - 06/2019 | Kragujevac, Serbia

PROJECTS

Haiku Clock (2026)

E-ink clock (MicroPython, Inky Frame with Raspberry Pi Pico) that shows a haiku for the current time, with NTP time sync and fallback servers, SD-card storage, and deep sleep between 15-minute updates to save power.

Music Recommendation System (2025)

Interactive song recommender (Python, Streamlit, scikit-learn) using KNN and cosine similarity over audio features, with PCA visualization of results.

Travel Agency (2023)

Web application (PHP/Laravel) for searching vacation destinations worldwide, with reservations and email notifications.